

Data Mining

Cluster Analysis: Advanced Concepts and Algorithms

Lecture Notes for Chapter 8

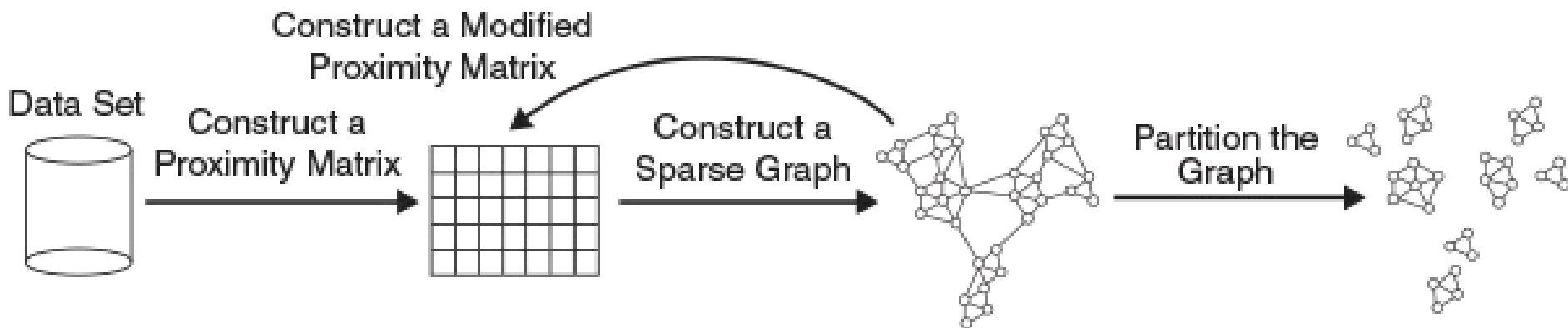
Introduction to Data Mining

by

Tan, Steinbach, Kumar

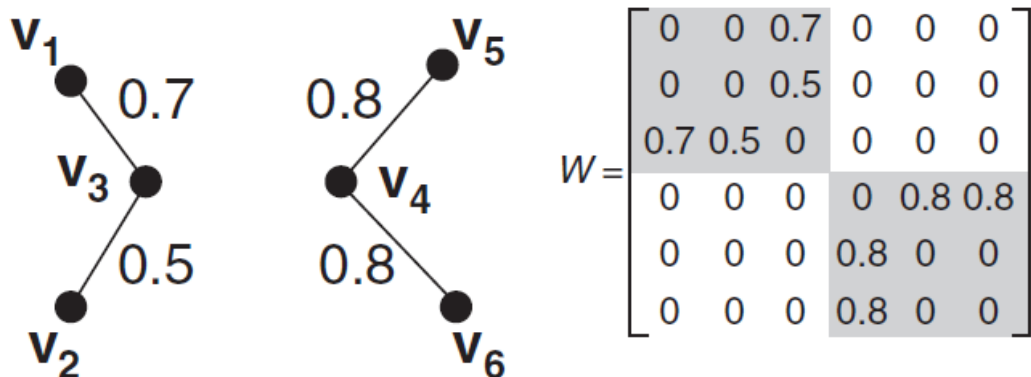
Outline

- Spectral Clustering
- Clustering Validity



Spectral Clustering

- Spectral clustering
 - is a **graph-based clustering** approach
 - Does a graph partitioning of the proximity graph of a data set
 - Breaks the graph into components, such that
 - ◆ The nodes in a component are strongly connected to other nodes in the component
 - ◆ The nodes in a component are weakly connected to nodes in other components
 - ◆ See simple example below ($\mathbf{W}$ is the proximity matrix)



Spectral Clustering

- For the simple graph below, the proximity matrix can be written as

$$\mathbf{W} = \begin{pmatrix} \mathbf{W}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_2 \end{pmatrix}$$

- Because the graph consists of two connected components finding clusters is easy.
- More generally, we need an automated approach
 - ◆ Must be able to handle graphs where the components are not completely separate
 - ◆ Spectral graph partitioning provides such an approach

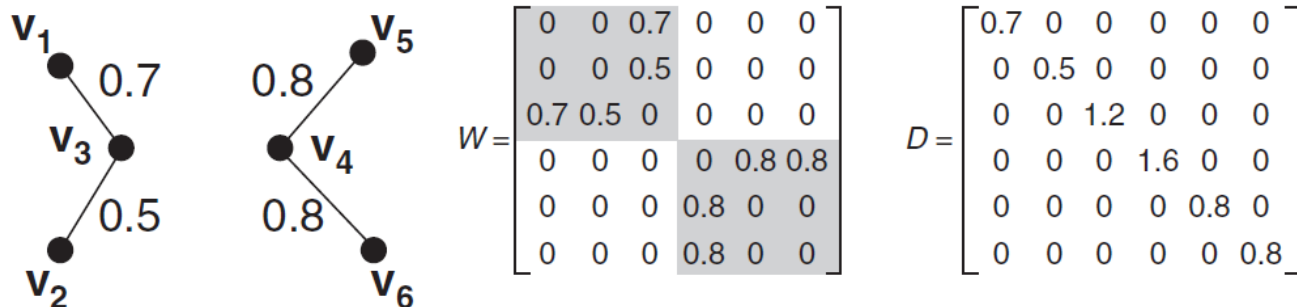
Clustering via Spectral Graph Partitioning

- Uses an eigenvalue based approach to do the graph partitioning
 - Based on the Laplacian matrix (**L**) of a graph, which is derived from the proximity matrix (**W**)
 - ◆ **W** is also known as the weighted adjacency matrix

- Define a **diagonal matrix D**

$$D_{ij} = \begin{cases} \sum_k W_{ik}, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases}$$

- k^{th} diagonal entry of **D** is the sum of the edges of the k^{th} node of **W**
 - See example below



Clustering via Spectral Graph Partitioning ...

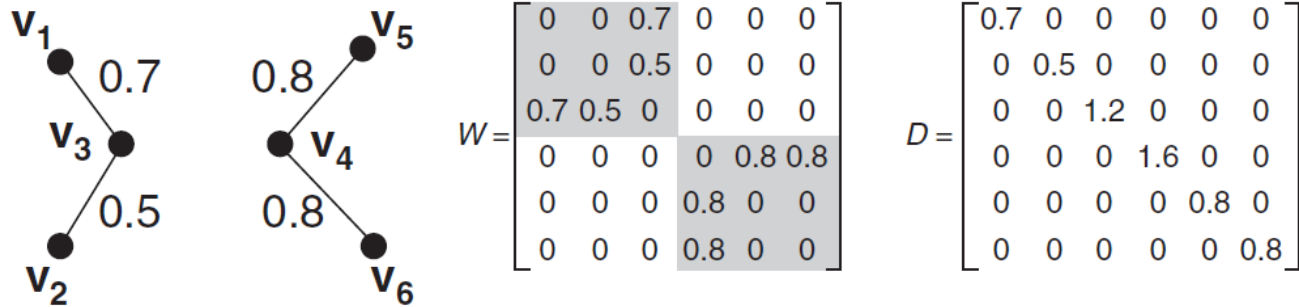
- Define a matrix called the Laplacian of the graph, L

$$L = D - W$$

- Can write L in terms of its eigenvalue decomposition as
 - $L = V\Lambda V^T$, where Λ is a diagonal matrix of the eigenvalues and V is the matrix of eigenvectors

$$L = \begin{bmatrix} 0.7 & 0 & -0.7 & 0 & 0 & 0 \\ 0 & 0.5 & -0.5 & 0 & 0 & 0 \\ -0.7 & -0.5 & 1.2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.6 & -0.8 & -0.8 \\ 0 & 0 & 0 & -0.8 & 0.8 & 0 \\ 0 & 0 & 0 & -0.8 & 0 & 0.8 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.58 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2.4 \end{bmatrix} \quad V = \begin{bmatrix} 0.58 & 0 & 0.64 & 0 & -0.50 & 0 \\ 0.58 & 0 & -0.76 & 0 & -0.31 & 0 \\ 0.58 & 0 & 0.11 & 0 & 0.81 & 0 \\ 0 & -0.58 & 0 & 0 & 0 & -0.82 \\ 0 & -0.5 & 0 & -0.71 & 0 & 0.41 \\ 0 & -0.5 & 0 & 0.71 & 0 & 0.41 \end{bmatrix}$$

Put it all together

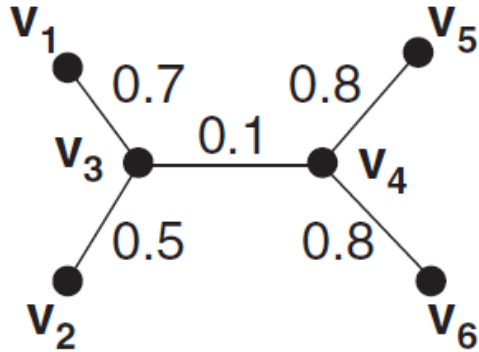


$$W = \begin{bmatrix} 0 & 0 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 & 0 \\ 0.7 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 & 0.8 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0.7 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.8 \end{bmatrix}$$

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A Slightly More Complicated Example



$$W = \begin{bmatrix} 0 & 0 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 & 0 \\ 0.7 & 0.5 & 0 & 0.1 & 0 & 0 \\ 0 & 0 & 0.1 & 0 & 0.8 & 0.8 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0.7 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.8 \end{bmatrix}$$

$$L = \begin{bmatrix} 0.7 & 0 & -0.7 & 0 & 0 & 0 \\ 0 & 0.5 & -0.5 & 0 & 0 & 0 \\ -0.7 & -0.5 & 1.3 & -0.1 & 0 & 0 \\ 0 & 0 & -0.1 & 1.7 & -0.8 & -0.8 \\ 0 & 0 & 0 & -0.8 & 0.8 & 0 \\ 0 & 0 & 0 & -0.8 & 0 & 0.8 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.06 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.58 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.80 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.88 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2.47 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.41 & -0.41 & 0.65 & 0 & 0.48 & -0.04 \\ 0.41 & -0.43 & -0.75 & 0 & 0.29 & 0.03 \\ 0.41 & -0.38 & 0.11 & 0 & -0.81 & 0.10 \\ 0.41 & 0.38 & 0 & 0 & 0.08 & -0.82 \\ 0.41 & 0.42 & 0 & -0.71 & 0.06 & 0.39 \\ 0.41 & 0.42 & 0 & 0.71 & 0.06 & 0.39 \end{bmatrix}$$

Spectral Graph Clustering Algorithm

- Given the Laplacian of a graph, it is easy to define a spectral graph clustering algorithm
- We simply apply k-means to the matrix consisting of the first k eigenvectors of L
- Note that we cluster the rows of that matrix

Algorithm 8.10 Spectral clustering algorithm.

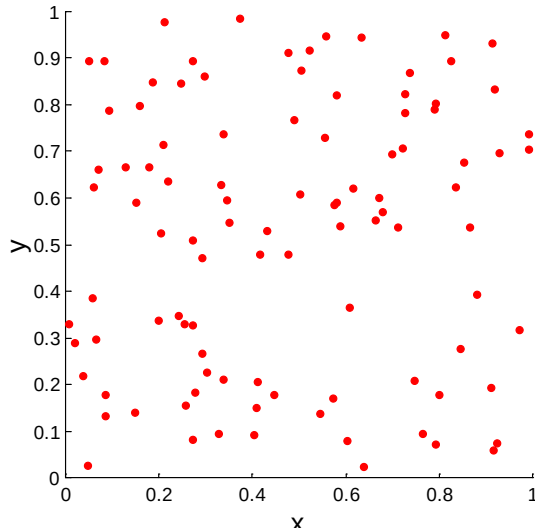
- 1: Create a sparsified similarity graph $\mathcal{G}$.
 - 2: Compute the graph Laplacian for $\mathcal{G}$, $\mathbf{L}$ (see Equation (8.20)).
 - 3: Create a matrix $\mathbf{V}$ from the first k eigenvectors of $\mathbf{L}$.
 - 4: Apply K-means clustering on $\mathbf{V}$ to obtain the k clusters.
-

Cluster Validity

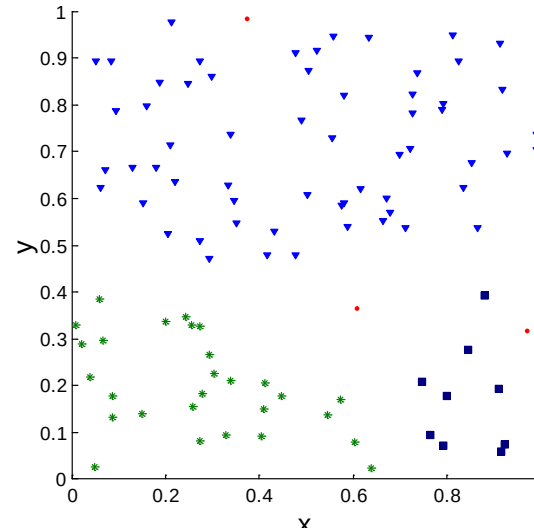
- For supervised classification we have a variety of measures to evaluate how good our model is
 - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
 - In practice, the clusters we find are defined by the clustering algorithm
- Then why do we want to evaluate them?
 - To avoid finding patterns in noise
 - To compare clustering algorithms
 - To compare two sets of clusters
 - To compare two clusters

Clusters found in Random Data

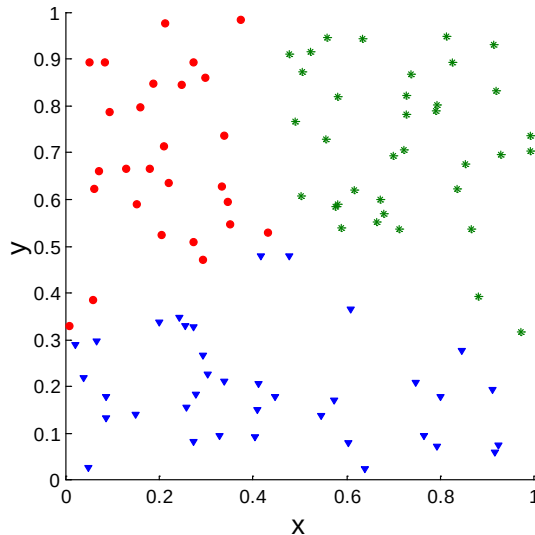
Random
Points



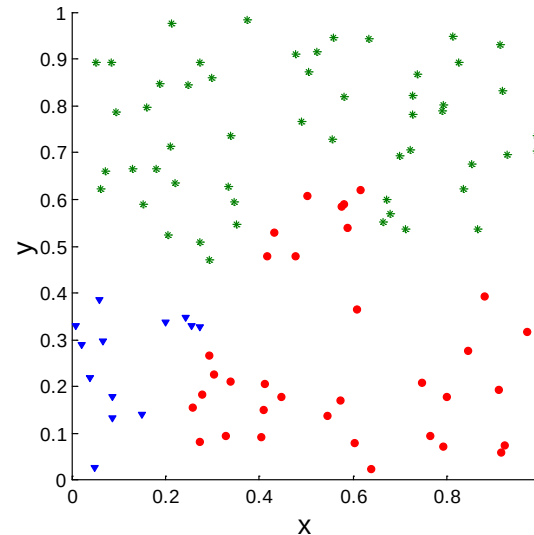
DBSCAN



K-means



Complete
Link



Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following two types.
 - **Supervised:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - ◆ Entropy
 - ◆ Often called *external indices* because they use information external to the data
 - **Unsupervised:** Used to measure the goodness of a clustering structure *without* respect to external information.
 - ◆ Sum of Squared Error (SSE)
 - ◆ Often called *internal indices* because they only use information in the data
- You can use supervised or unsupervised measures to compare clusters or clusterings

Supervised Measures of Cluster Validity: Entropy and Purity

- **Entropy:** The degree to which each cluster consists of objects of a single class.

p_{ij} : This is the probability that a member of cluster i belongs to class j . m_{ij} is the number of objects in cluster i that belong to class j , m_i is the total number of objects in cluster i .

$$p_{ij} = m_{ij}/m_i$$

e : This is the overall entropy, which is a weighted average of the entropies of all clusters.
 m is the total number of objects across all clusters,
 e_i is the entropy for each cluster i .

$$e = \sum_{i=1}^K \frac{m_i}{m} e_i$$

e_i : This is the entropy for cluster i , measuring the degree to which the cluster is mixed with different classes.

$$e_i = - \sum_{j=1}^L p_{ij} \log_2 p_{ij}$$

L represents the number of unique classes within the dataset.
 K represents the number of clusters that the data is divided into.

Table 5.9. K-means Clustering Results for LA Document Data Set

Cluster	Entertainment	Financial	Foreign	Metro	National	Sports	Entropy	Purity
1	3	5	40	506	96	27	1.2270	0.7474
2	4	7	280	29	39	2	1.1472	0.7756
3	1	1	1	7	4	671	0.1813	0.9796
4	10	162	3	119	73	2	1.7487	0.4390
5	331	22	5	70	13	23	1.3976	0.7134
6	5	358	12	212	48	13	1.5523	0.5525
Total	354	555	341	943	273	738	1.1450	0.7203

Supervised Measures of Cluster Validity: Entropy and Purity

- **Purity:** the extent to which a cluster contains objects of a single class.
- For each cluster, the class distribution of the data is calculated first, i.e., for cluster i we compute the probability that a member of cluster i belongs to class j

$$p_{ij} = m_{ij}/m_i$$

$$purity = \sum_{i=1}^K \frac{m_i}{m} purity(i).$$

$$purity(i) = \max_j p_{ij}.$$

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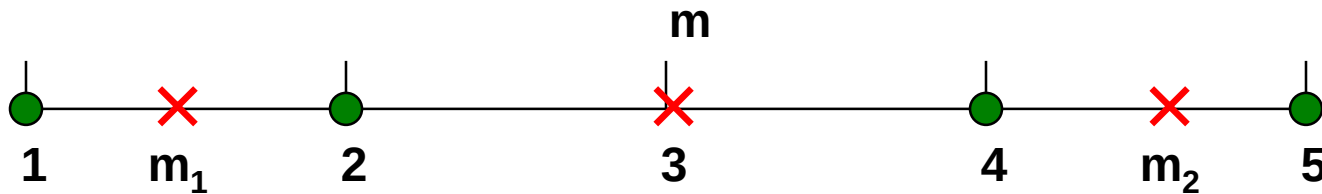
Unsupervised Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster
 - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
 - Cohesion is measured by the within cluster sum of squares (SSE)
$$SSE = \sum_i \sum_{x \in C_i} (x - m_i)^2$$
 - Separation is measured by the between cluster sum of squares
$$SSB = \sum_i |C_i| (m - m_i)^2$$

Where $|C_i|$ is the size of cluster i

Unsupervised Measures: Cohesion and Separation

- Example: SSE
 - $SSB + SSE = \text{constant}$



K=1 cluster: $SSE = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$

$$SSB = 4 \times (3 - 3)^2 = 0$$

$$Total = 10 + 0 = 10$$

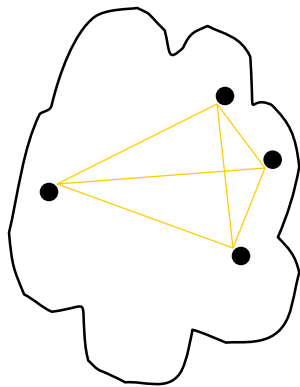
K=2 clusters: $SSE = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$

$$SSB = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

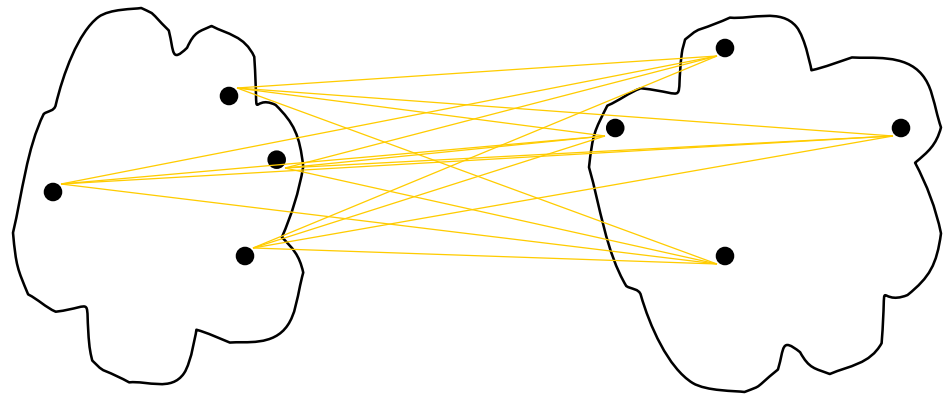
$$Total = 1 + 9 = 10$$

Unsupervised Measures: Cohesion and Separation

- A proximity graph-based approach can also be used for cohesion and separation.
 - Cluster cohesion is the sum of the weight of all links within a cluster.
 - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion

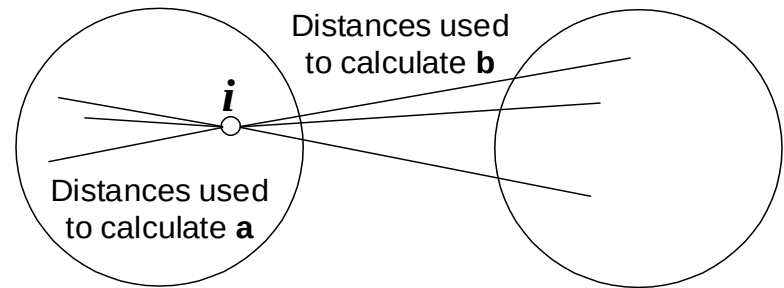


separation

Unsupervised Measures: Silhouette Coefficient

- Silhouette coefficient combines ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings
- For an individual point, i
 - Calculate a = average distance of i to the points in its cluster
 - Calculate b = average distance of i to points in another cluster
 - The silhouette coefficient for a point is then given by

$$s = (b - a) / \max(a, b)$$



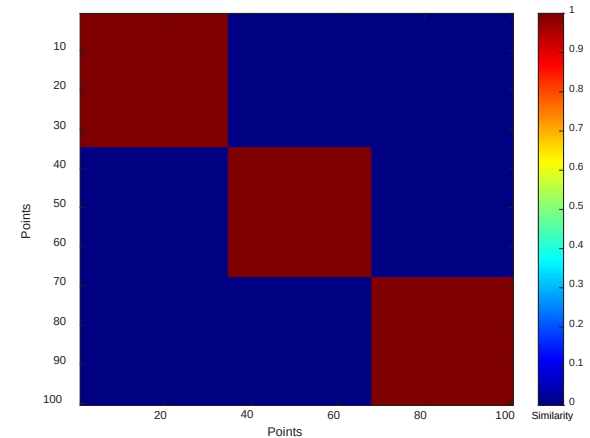
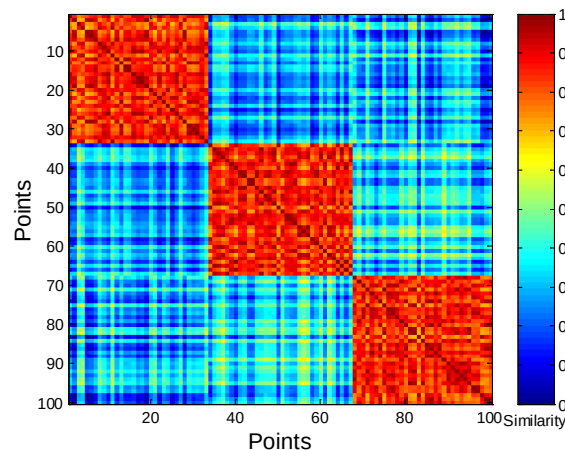
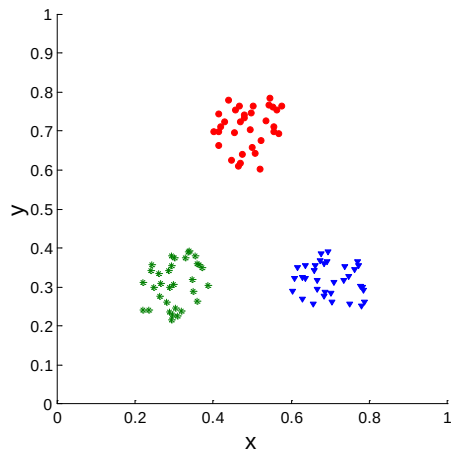
- Value can vary between -1 and 1
 - Typically ranges between 0 and 1.
 - The closer to 1 the better.
- Can calculate the average silhouette coefficient for a cluster or a clustering

Measuring Cluster Validity Via Correlation

- Two matrices
 - Proximity Matrix
 - Ideal Similarity Matrix
 - ◆ One row and one column for each data point
 - ◆ An entry is 1 if the associated pair of points belong to the same cluster
 - ◆ An entry is 0 if the associated pair of points belongs to different clusters
- Compute the correlation between the two matrices
 - Since the matrices are symmetric, only the correlation between $n(n-1) / 2$ entries needs to be calculated.
- High magnitude of correlation indicates that points that belong to the same cluster are close to each other.
 - Correlation may be positive or negative depending on whether the similarity matrix is a similarity or dissimilarity matrix
- Not a good measure for some density or contiguity based clusters.

Measuring Cluster Validity Via Correlation

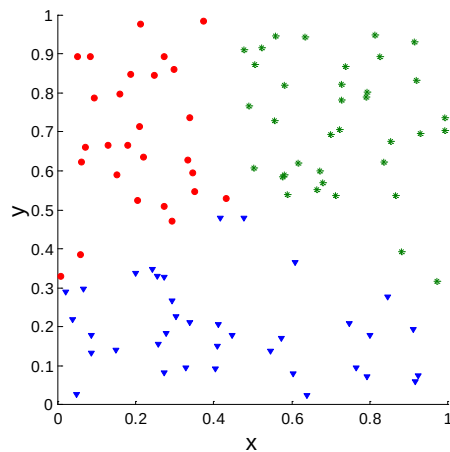
- Correlation of ideal similarity and proximity matrices for the K-means clustering of the following well-clustered data set.



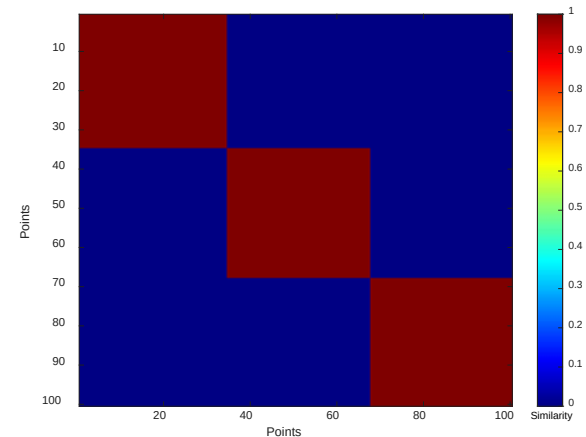
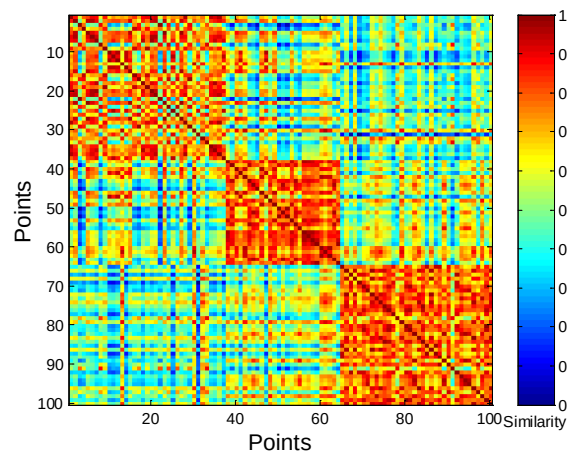
Corr = 0.9235

Measuring Cluster Validity Via Correlation

- Correlation of ideal similarity and proximity matrices for the K-means clustering of the following random data set.



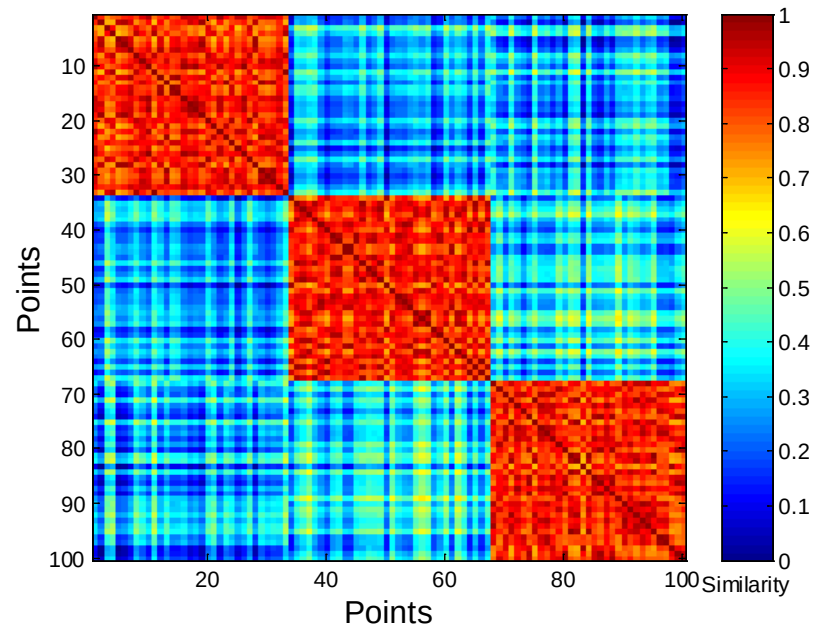
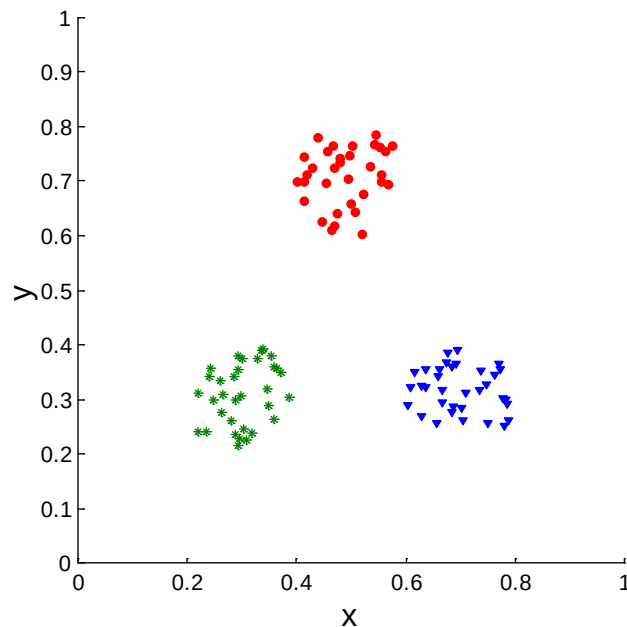
K-means



Corr = 0.5810

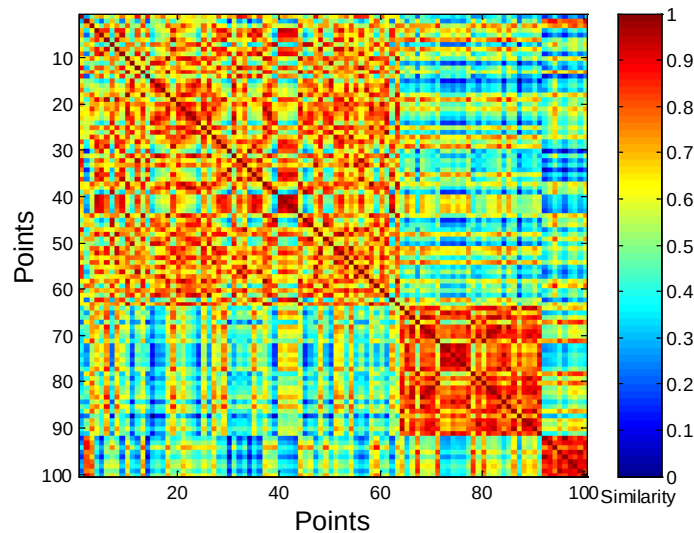
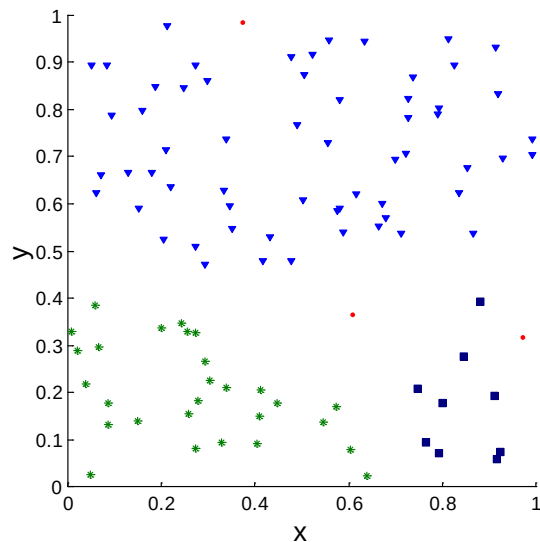
Judging a Clustering Visually by its Similarity Matrix

- Order the similarity matrix with respect to cluster labels and inspect visually.



Judging a Clustering Visually by its Similarity Matrix

- Clusters in random data are not so crisp



DBSCAN

Final Comment on Cluster Validity

“The validation of clustering structures is the most difficult and frustrating part of cluster analysis.

Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage.”

Algorithms for Clustering Data, Jain and Dubes

- H. Xiong and Z. Li. *Clustering Validation Measures*. In C. C. Aggarwal and C. K. Reddy, editors, *Data Clustering: Algorithms and Applications*, pages 571–605. Chapman & Hall/CRC, 2013.